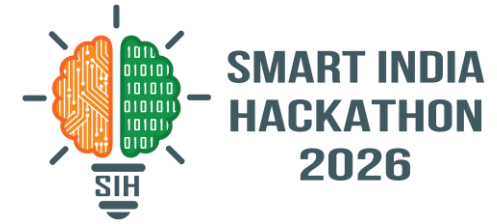
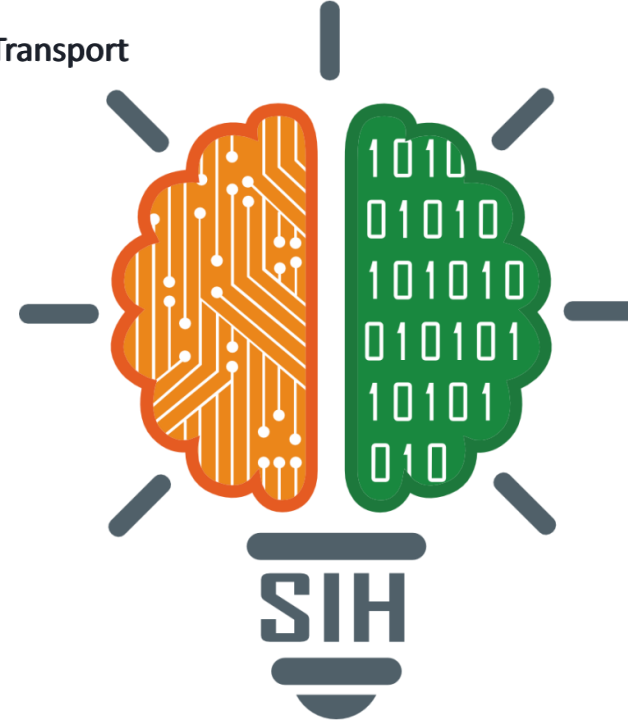


SMART INDIA HACKATHON 2026

IDEA SUBMISSION — TITLE PAGE



- **Problem Statement ID:** [SIH26124](#)
- **Problem Statement Title:** AI-Powered Mobile Urban Intelligence Platform Using Public Transport Fleet
- **Theme:** Smart Automation
- **PS Category:** Software
- **Organization:** Bharat Electronics Limited (BEL)
- **Team ID:** [SIH26-LBS-007](#)
- **Team Name (Registered):** **The Spartiates**
- **Idea / Project Title:** CityFleet AI — Mobile Urban Intelligence Platform



PROPOSED SOLUTION (IDEA OVERVIEW)

- **Distributed Mobile Sensing Grid:** Turns scheduled public transit buses into an autonomous city-wide road and traffic sensing network without deploying expensive stationary roadside sensors.
- **Edge-to-Cloud Intelligence:** Bus-mounted cameras perform real-time edge defect detection, streaming lightweight JSON summaries (~2 KB) to a central municipal GIS command platform.

HOW IT ADDRESSES THE PROBLEM

- **Continuous & Proactive Auditing:** Replaces subjective, delayed citizen grievance reports with automated, timestamped, GPS-verified road health surveys.
- **Zero Infrastructure Capex:** Operates directly on existing public bus fleets, saving over 85% in road survey costs compared to dedicated inspection vehicles.

INNOVATION & UNIQUENESS OF THE SOLUTION

- **Multi-Bus Spatial-Temporal Fusion:** Cross-validates defect coordinates across multiple buses, escalating event confidence from 88% to 99.8% while suppressing false alarms.
- **Dynamic Road Health Index (RHI: 0–100):** Computes an objective, segment-wise degradation score (0–100) to auto-prioritize municipal repair tenders based on traffic exposure.

TECHNOLOGIES TO BE USED (TECH STACK)

- **Edge Compute & AI:** Python 3.11, OpenCV, YOLOv8/v11 (ONNX & TensorRT runtime at 30+ FPS), ByteTrack multi-object tracker, local SQLite buffer queue.
- **Backend & Geospatial Database:** FastAPI (Async REST & WebSockets), PostgreSQL 16 + PostGIS (spatial indexing & OSM road-snapping), Redis 7 (event broker), MinIO (S3 evidence crops).
- **GIS Command Center UI:** React 18, TypeScript, Vite, MapLibre GL JS / Deck.gl vector mapping, Tailwind CSS, Chart.js municipal analytics.

METHODOLOGY & IMPLEMENTATION PROCESS

- **1. Edge Perception & Anonymization:** Dashcam video is processed on-device; detects potholes, cracks, ravelling, waterlogging, and counts traffic while blurring PII (faces/license plates).
- **2. Offline-Resilient Sync:** Lightweight JSON events (~2 KB) are transmitted via TLS MQTT; stored in local SQLite when traversing tunnels/dead zones and synced on reconnection.
- **3. Multi-Bus Fusion & Decision Engine:** PostGIS clusters sightings within 15m radius, applies Bayesian consensus confidence $[C_{\text{fused}} = 1 - \prod(1 - w_i \cdot c_i)]$, and computes maintenance priority scores.

ANALYSIS OF FEASIBILITY (TECHNICAL, OPERATIONAL & ECONOMIC)

- **Technical Feasibility:** YOLOv8/11 models achieve 30+ FPS on edge hardware (Jetson / Mini-PC); validated on multi-national 47K+ road damage benchmark (RDD2022).
- **Operational Viability:** Zero route disruption; operates non-intrusively on scheduled municipal buses during normal transit without driver interaction.
- **Economic Viability:** Avoids expensive dedicated survey vehicle tenders; early micro-repairs save up to 10x in structural road reconstruction costs.

POTENTIAL CHALLENGES, RISKS & MITIGATION STRATEGIES

- **Challenge 1 — GPS Drift in Urban Canyons:** Mitigated via PostGIS geodesic road-snapping to OpenStreetMap (OSM) vector networks + 15m radius spatial clustering buffer.
- **Challenge 2 — Cellular Dead Zones / Tunnels:** Mitigated via edge SQLite circular store-and-forward buffer; zero telemetry loss during network dropouts.
- **Challenge 3 — Single-Camera False Positives:** Mitigated via 3-frame edge temporal persistence filter + central multi-bus Bayesian consensus fusion across distinct passes.
- **Challenge 4 — Privacy & PII Regulations:** Mitigated via privacy-by-design: automatic on-device Gaussian blurring of faces and license plates prior to transmission.

POTENTIAL IMPACT ON TARGET AUDIENCE (MUNICIPALITIES, COMMUTERS & BEL)

- **Municipal Governance:** Replaces reactive citizen complaints with audited, timestamped, GPS-verified road health work orders and automated contractor SLA tracking.
- **Public Safety & Commuters:** Addresses India's critical road safety challenge (4,000+ annual pothole fatalities); eliminates accident hazards and monsoon waterlogging bottlenecks.
- **BEL & Strategic Dual-Use:** Directly integrates into Bharat Electronics Limited Smart City ICCC platforms; dual-use applicability for military convoy route clearance.

DIRECT & QUANTIFIABLE BENEFITS

- **Economic & Financial:** >85% cost reduction in road survey operations; timely micro-repairs prevent costly full-depth road reconstruction (\$1 vs \$10 repair rule).
- **Citizen Vehicle Savings:** Significant reduction in tire blowouts, suspension damages, and vehicular wear-and-tear for urban commuters.
- **Environmental & Traffic Flow:** Prevents traffic slowdowns caused by bad road patches, reducing commuter idle time and urban carbon emissions.

BENCHMARK DATASETS & TECHNICAL STANDARDS

- **Road Damage Dataset (RDD2022):** Global multi-national benchmark dataset containing 47,000+ annotated road distress images across 6 countries (IEEE BigData / CVPR).
- **OpenStreetMap (OSM) Vector Road Topology:** Geodesic network topology dataset utilized for road-segment snapping and traffic attribution.
- **Indian Roads Congress (IRC:SP:19 & IRC:82):** Standards for maintenance and classification of asphalt/concrete road distresses.
- **MoRTH India Annual Road Accident Reports:** Ministry of Road Transport & Highways empirical causality and infrastructure defect data.

PEER-REVIEWED ACADEMIC CITATIONS

- **1. Arya, D. et al. (2022):** "Global Road Damage Detection: State-of-the-Art Deep Learning Models and Benchmark Dataset" — IEEE Trans. on Intelligent Transportation Systems.
- **2. Ester, M. et al. (1996):** "A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise (ST-DBSCAN)" — Proc. of KDD.
- **3. Ultralytics (2024):** "YOLOv8 & YOLOv11 Real-Time Object Detection and Edge Acceleration Architecture" — TensorRT & ONNX deployment pipelines.
- **4. Zhang, Y. et al. (2022):** "ByteTrack: Multi-Object Tracking by Associating Every Detection Box" — European Conference on Computer Vision (ECCV).
- **5. PostGIS & OGC Guidelines:** "Spatial and Geographic Objects for PostgreSQL Systems" — Open Geospatial Consortium (OGC) specifications.